

TEACHING PHILOSOPHY STATEMENT

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I decided to pursue a PhD because I want to be a professor. Through my numerous teaching experiences, my idea about what it means to be a professor has expanded dramatically. Beyond the role of knowing mathematical subjects and explaining them in lectures, I now understand and practice that college involves complex works of planning lessons and assignments to fulfill the learning objectives each week and creating multiple strategies to explain the intuitive ideas behind mathematical concepts to students with multiple backgrounds. It includes supporting my students, not only academically but beyond the classroom. It encompasses building a community to support educators and learners. Currently, being a mathematics professor means being a community builder who fosters students' learning of mathematics through a holistic approach to the whole person. I am confident that over the years the meaning of being a professor will continue to grow.

The **pedagogy of care** is the cornerstone of my teaching philosophy. The baseline of my role as their instructor is *helping my students inside and outside the classroom*. Pedagogy of care involves the acknowledgment that my students have a life filled with responsibilities and interests outside academia and this pedagogy manifests both in academic and personal support. For example, if a student faces adverse circumstances that prevent them from fulfilling some assignments, I will set alternative deadlines to allow them to make up the work. In a similar way, I take 3 minutes at the beginning of each class to encourage them to live a well-rounded life through motivational, challenging, or thought-provoking ideas. In this way, I demonstrate that I care about their well-being as whole individuals. When I was an undergraduate student, I felt that I received personalized instruction in many ways, as each of my professors knew my learning process firsthand by actively guiding all classroom activities. As an educator, I want to create a personalized experience for my students and make them feel I care about them individually.

Pedagogy of care also means *accompanying my students in their mathematical learning process*. I appreciate it when a professor helps me comprehend mathematical concepts in a way that enhances what I am able to learn alone with a textbook (and other resources). The textbook for a student is like a map for a sailor. The map contains all the information about reaching the destination. However, only someone who learns to be a navigator and read maps can extract the route and take the ship to the desired place. Similarly, as my professors did, my objective is “preparing my students to be navigators” by unveiling the intuitive ideas that underlie mathematical concepts and sharing with them the clever solving techniques and mnemonics I know and enjoy.

Following the analogy, I was a lone navigator in my undergraduate studies, solving problems on my own and rarely encouraged to engage in group work. Reflecting on this experience, I now recognize the value of *collaborative learning, which I consider central to my classroom*. From the very first day, I motivate my students to work in groups. I remind them that we, as humans, flourish when we share with one another, and learning mathematics is no exception. Groupwork allows everyone to thrive: those who are unsure have the chance to ask and feel supported by a peer, while those who explain have the opportunity to articulate mathematical concepts in their own words, thereby making the knowledge their own.

However, in my days as a lone navigator I recognized that *learning mathematics is based on doing*. I tell my students that mathematics is like a sport for our brain. If we want to play it well, we need multiple hours of practice: the more we do it, the better we get. Hence, I utilize quizzes, problem-solving sessions, and suggested practice problems to encourage my students to actively engage with the material, deepen their understanding and ultimately make that knowledge their own. Ironically, when I was studying, most of the time I did not have quizzes nor homework, only exams. On one hand, solving problems became my natural way to internalize concepts and not an obligation to achieve a good grade. Looking forward to inspiring the same appreciation from my students, I evaluate homework based more on effort than results. My assessment is focused on providing feedback to enhance their understanding.

On the other hand, I recognize that I was assessed only with exams because my professors did not have the time, nor the assistance or technology to carry out other assignments. I am grateful for technology that makes grading more efficient. Nevertheless, beyond making grading easier, technology offers opportunities to support students' learning. During my research project "Students' perspectives of the use of Khanmigo as a mathematics AI tutor", I gained first-hand insight into how students engage with generative AI tools when learning mathematics. This experience reinforced the importance of thoughtfully integrating AI into my classroom to promote its clever and effective use. At the same time, I strive to help students cultivate self-awareness in their learning, recognizing when they need to push themselves further in solving a problem or seeking guidance from tutors and peers; and when it is appropriate to turn to AI for support. Just as I aim to support my students through personal attention and guidance, I also embrace technology to promote their mathematical understanding and independence. In this way, they become navigators on their own.

Finally, my teaching philosophy has been enriched not only by my students but also the mentorship and collaboration of my supervisors and colleagues throughout my PhD, and I hope to have contributed to their growth in turn. Over the past few years, I have had the opportunity to observe and give constructive feedback to many of my fellow doctoral students, which has both helped me build the teaching community I value so much, along with enhancing my own teaching practice by learning from my peers and their creative teaching methods. I was also selected to facilitate the year-long orientation program for incoming graduate students, which involved both planning and leading teaching orientation seminars that are designed to support their transition into the teaching academic environment of the university and department. This orientation addresses both classroom teaching techniques and mindset/community aspects of teaching, which has honed my own practice and philosophy. I am grateful for these opportunities to share my teaching experiences with my peers. These collaborations have helped me to recognize that **teaching mathematics is a collective endeavor** where care, teamwork and shared knowledge benefit the whole mathematical community. It is a ship taken care of by all. I look forward to continuing to nurture my students' mathematical learning, share my experiences, and contribute to building a community where students and educators can thrive.